PsyNamic: A dynamic and comprehensive systematic review of psychedelic therapy in psychiatric disorders

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Background and research questions

The resurgence of interest in the therapeutic potential of psychedelic substances for psychiatric disorders has heralded a new era of clinical research. Substances such as LSD, psilocybin, mescaline, DMT, 5-MeO-DMT, MDMA, ketamine, ibogaine, and salvinorin A are being investigated for their efficacy in treating conditions like eating disorders, depression, and anxiety. However, the rapid accumulation of clinical trial data presents a challenge for manual synthesis and the continuous updating of evidence. This study aims to address this gap by establishing a living systematic review that dynamically incorporates new research findings on the use of psychedelic substances in psychiatric treatment. Utilizing natural language processing (NLP) techniques, including large language models, we plan to automate the classification of studies and extraction of relevant data. Our approach will enable the real-time synthesis of evidence, ensuring that the most current data is readily accessible. The outcomes of this research will be freely available in an online data warehouse, providing a comprehensive and up-to-date resource for both clinicians planning future trials and interested laypersons seeking evidence-based information on psychedelic therapy. This endeavor promises to enhance the efficiency of evidence synthesis in the burgeoning field of psychedelic research and offer insights for the evidence-based application of psychedelic therapy in psychiatric care for lay people.

Study Information

Hypotheses

No hypotheses are formulated. This is a systematic review.

Design Plan

Study type

Meta-Analysis - A systematic review of published studies.

Blinding

No blinding is involved in this study.

Is there any additional blinding in this study?

Not applicable.

Study design

Systematic review (and potentially meta-analysis).

Randomization

Not applicable.

Sampling Plan

Existing Data

Registration prior to accessing the data.

Explanation of existing data

Preliminary literature searches have been performed on PubMed to probe the available evidence.

Data collection procedures

<u>Identify relevant studies using a comprehensive search string</u>

An information specialist of the University of Zurich conducted a comprehensive literature search in Embase, Medline via PubMed, Cochrane Trials (CENTRAL), PsycInfo, Scopus, and Web of Science.

Search

LSD, Lysergic acid diethylamide, delysid, LSD-25

Psilocybin, 4-PO-DMT

Psilocin, 4-HO-DMT

Mescaline, mescalin, 3,4,5-trimethoxyphenethylamine, 3,4,5-Trimethoxyphenethylamine, TMPEA, Peyote

DMT, N,N-Dimethyltryptamine, N,N-DMT

5-MeO-DMT, 5-methoxy-N,N-dimethyltryptamine, O-methyl-bufotenin

3,4-Methylenedioxymethamphetamine, MDMA, ecstasy

2C-B, 2C-I, 2C-E, 2C-C

Ketamine, ketamin

Ibogaine

Salvia divinorum

See **comprehensive search string** for details.

Inclusion and exclusion criteria

We will **include** all clinical studies which report on the **use of psychedelic substances (see below) in human populations**, this includes:

-Case reports/case series with only 1/few included individuals.

-Cohort studies, including studies observing the use of psychedelic substances within community-dwelling individuals; also including studies assessing potential (long-term) adverse effects of psychedelics.

-Case-control studies.

-Randomized-controlled trials.

-Any other clinical study type, including e.g., pharmacological interaction studies or metabolite studies (as long as they involve human subjects).

Of note, participants do NOT need to be patients, i.e., recreational use of psychedelic substances is also included.

We will also **include** the following study types:

-systematic reviews/meta-analysis: mentioning systematic review/meta-analysis or, if not specifically declared as systematic review, reporting at least two biomedical databases in the abstract (e.g., PubMed and EMBASE).

-Study protocols, including clinical study registries.

-Studies which re-analyze data from former clinical studies, e.g., pooling data from two RCTs.

-Corrections of articles related to included article types.

We will <u>include</u> the following substances:

Name	**Synonyms and/or psychoactive metabolites (NOT comprehensive)**
LSD	Lysergic acid diethylamide, delysid, LSD-25, acid (**There are many more synonyms**)
Psilocybin	4-PO-DMT, psilocin, 4-HO-DMT, magic mushrooms
Mescaline	mescalin, 3,4,5-trimethoxyphenethylamine, 3,4,5-Trimethoxyphenethylamine, TMPEA, Peyote, San Pedro
DMT (including ayahuasca)	N,N-Dimethyltryptamine, N,N-DMT
5-MeO-DMT	5-methoxy-N,N-dimethyltryptamine, O-methyl-bufotenin, bufetonin, Bufotenine, N,N-Dimethylserotonin, N,N-Dimethylserotonin, N,N-Dimethyl-5-hydroxytryptamine, Mappin, Mappine, Dimethylserotonin, Cohoba, DM5-HT, 5-Hydroxy-N,N-dimethyltryptamine, More synonyms on: https://pubchem.ncbi.nlm.nih.gov/compound/Bufotenine#section=Depositor-Supplied-Synonyms
MDMA	3,4-Methylenedioxymethamphetamine, Molly, Ecstasy
Ketamine	ketamin
Ibogaine	Tabernanthe iboga, Noribogaine, O-desmethylibogaine, 12-hydroxyibogamine, Iboga, noribogaine
Harmin	Harmine, Harmaline, banisterin, banisterine, telopathin, telepathine, leucoharmine, yagin, yageine
Salvia divinorum	Salvinorin A

We will **exclude** the following study types:

-Studies not employing psychedelic substances, i.e., THC, CBD, and other derivates. Other illicit substances such as cocaine, heroine, etc.

-Philosophical discussions about psychedelic use

-Animal studies

-(Non-systematic) reviews, including opinions, commentaries, editorials, etc.

-Grey literature including conference abstracts

-no abstract available

Sample size

Not applicable.

Sample size rationale

Not applicable.

Stopping rule

Not applicable.

Variables

Manipulated variables

Not applicable.

Measured variables

1) Study screening

References will be screened in duplicate by two independent reviewers (DB, BH) in the app ASReview. Discrepancies will be resolved by a third reviewer (BVI) and in discussion with DB and BH.

Benchmark studies to include in ASReview

Ketamine Versus Midazolam for Recurrence of Suicidality in Adolescents

Acute effects of MDMA and LSD co-administration in a double-blind placebo-controlled study in healthy participants

Trial of psilocybin versus escitalopram for depression

Benchmark studies to exclude in ASReview

Nootropic effects of LSD: Behavioral, molecular and computational evidence

Psilocybin in Palliative Care: An Update

Δ(9)-Tetrahydrocannabinol (THC) impairs visual working memory performance: a randomized crossover trial

ASReview sorting will be stopped after 20 irrelevant records in a row.

Additional benchmark studies to include

https://doi.org/10.1038/s41591-023-02455-9

https://doi.org/10.1016/S2215-0366(16)30065-7

https://doi.org/10.1038/s41591-023-02565-4

https://doi.org/10.1038/s41386-023-01609-0

https://doi.org/10.1038/s41386-022-01297-2

https://doi.org/10.1038/npp.2016.82

https://doi.org/10.1002/cpt.2958

https://doi.org/10.1007/s13318-023-00822-y

Priority for exclusion:

1) A paper outside the realm of psychedelic research.

2) Non-systematic review.

3) Animal studies of psychedelic research (studies including humans and animals will be included).

Included references will be exported from ASRevuew to an Endnote library. Full text PDFs will be fetched by the endnote fetch tool, complemented by manual fetching of PDFs.

Data extraction

To train a model to classify studies and automated data extraction, a random sample of studies will be annotated manually by two independent reviewers in Prodigy. The following data will be extracted by two independent reviewers on abstract level (no full text data extraction will be conducted):

We defined a comprehensive list of parameters to be extracted from the abstracts of interest (see **annotation guidelines**). Briefly, these parameters relate to the study characteristics (e.g., study 1) Study type: E.g., randomized-controlled trial (RCT), case report or case series, systematic review/meta-analysis, study on recreational drug use, study protocols, re-analysis

2) Substance: LSD, psilocybin, mescaline, DMT, 5-MeO-DMT, MDMA, ketamine, ibogaine, and salvinorin A (or chemically related/similar substances; can be more than one substance)

3) Application: Relates to the disease being treated with the psychedelic substance, e.g., anxiety disorder, eating disorder, depression, posttraumatic stress disorder (could also be recreational use or healthy individuals for e.g., pharmacokinetics).

4) Number of participants: as class (<10, 11-50, >50)

5) Sex of participants: male, female, both, not reported

6) Clinical trial phase: 0, 1, 2, 3, 4.

7) Adverse event(s): No adverse event(s) reported, adverse event(s) reported, no reporting of adverse event(s).

8) Conclusion: positive, neutral, negative, mixed (usually last sentence of abstract, sentiment analysis).

In addition, we will use the following metadata: doi, title, authors, publication year, journal, and number of authors.

Based on this corpus of manually annotated titles and abstracts, we will develop/train/fine-tune natural language processing approaches such as regular expressions (RegEx) and (large) language models (such as BERT-based models or GPT-based models) which aim at automatically extracting these information from the remaining, non manually annotated abstracts. Performance of models will be measured using F1-scores, precision, and recall.

Indices

Not applicable.

Analysis Plan

Statistical models

Descriptive statistics and visualization for each substance, study type, and disease.

Transformations

Not applicable.

Inference criteria

Not applicable.

Data exclusion

Not applicable.

Missing data

Not applicable.

Exploratory analysis

Not applicable.

Other

Other

All data will be visualized in dyamic fashion in a free online data warehouse called PsyNamic.